

Lecture no 21

Reduction of Order :-

Ex 6.3

$$y'' - 25y = 0 \quad (1)$$

$$y = e^{5x}$$

Let $y_2 = u \cdot y_1$ is other solution $\therefore u = u(x)$
 $y_1 = y_1(x)$

$$y_2 = u(x) e^{5x}$$

$$y_1' = u(x) 5e^{5x} + u'(x) e^{5x}$$

$$y_2'' = 5[u(x) \cdot 5e^{5x} + u'(x) e^{5x}] + u''(x) e^{5x} + 5u'(x) e^{5x}$$

$$= 25u(x) e^{5x} + 5u'(x) e^{5x} + u''(x) e^{5x} + 5u'(x) e^{5x}$$

$$= 25u(x) e^{5x} + 10u'(x) e^{5x} + u''(x) e^{5x}$$

put y and y'' in (i)

$$= 25u(x) e^{5x} + 10u'(x) e^{5x} + u''(x) e^{5x} - 25u(x) e^{5x} = 0$$

$$\Rightarrow 10u'(x) e^{5x} + u''(x) e^{5x} = 0$$

$$e^{5x} (10u'(x) + u''(x)) = 0 \Rightarrow 10u'(x) + u''(x) = 0$$

$$w(x) = u'(x)$$

req(ii)

$$w'(x) = u''(x)$$

$$w'(x) + 10w(x) = 0$$

$$\frac{dw}{dx} + 10w = 0 \Rightarrow \frac{dw}{w} + 10w = 0$$

$$\Rightarrow \frac{dw}{w} = -10 dx$$

$$\int \frac{dw}{w} = -10 \int dx$$

$$\ln|w| = -10x + C$$

$w =$ Anti log:-

$$w = e^{-10x+C}$$

$$w = Ce^{-10x}$$

$$u'(x) = Ce^{-10x}$$

$$\frac{du}{dx} = Ce^{-10x}$$

$$du = d(Ce^{-10x})$$

Integrate b.s:-

$$u = \frac{Ce^{-10x}}{-10} + C_1$$

$$u = C_1 e^{-10x} + C_2$$

$$y_2 = (C_1 e^{-10x} + C_2) e^{5x}$$

$$y_2 = C_1 e^{-5x} + C_2 e^{5x}$$

$$y_2 = e^{-5x}$$

$$y_2 = e^{5x}$$

General Solution

$$y = C_1 e^{-5x} + C_2 e^{5x}$$

Formula :-

$$y''(x) + p(x)y'(x) + q(x) = 0$$

y_1 is a solution

then

$$y_2 = u \cdot y_1$$

$$u = \int \frac{e^{-\int p(x) dx}}{y_1^2} dx$$

$$p(x) = 0, y_1 = e^{5x}$$

$$= \int \frac{e^{-\int 0 dx}}{(e^{5x})^2} dx$$

$$u = \int \frac{e^{-0}}{e^{10x}} dx$$

$$u = e^{-10x}$$

$$\begin{aligned} y_2 &= u \cdot y_1 \\ &= e^{-10x} \cdot e^{5x} \\ &= e^{-5x} \end{aligned}$$

Q.93

$$x^2 y'' - 7xy' + 16y = 0 ; \quad y_1 = x^4$$

Taking dividing by x^2

$$y'' - \frac{7xy'}{x^2} + \frac{16y}{x^2} = 0$$

$$y'' - 7\frac{y'}{x} + \frac{16y}{x^2} = 0$$

$$P(x) = \frac{-7}{x}$$

~~$u = e^{-\int P(x) dx}$~~

$$u = \int \frac{e^{-\int P(x) dx}}{y_1^2} dx$$

$$u = \int \frac{e^{-\int (-7/x) dx}}{(x^4)^2} dx$$

$$u = \int \frac{x^7}{(x^4)^2} dx$$

$$u = \int \frac{x^7}{x^8} dx$$

$$u = \int \frac{1}{x} dx$$

$$u = \ln x$$

$$y_2 = u \times y_1 = x^4 \ln x$$

S.D:-

$$y = C_1 x^4 + C_2 x^4 \ln x$$

SO.33

$$y'' + 16y = 0 ; \quad y_1 = \cos 4x$$

$$\begin{aligned} p(x) = 0 \text{ so } u &= \int \frac{1}{y_1^2} \\ &= \int \frac{1}{\cos^2 4x} \\ &= \int \sec^2 4x \\ &= \frac{\tan 4x}{4} \end{aligned}$$

$$y_2 = u \times y_1 = \frac{\cos 4x \tan 4x}{4}$$

~~$y_1 \times u = y_2$~~

$$y = C_1 \cos 4x + C_2 \frac{\cos 4x \tan 4x}{4}$$

$$y = C_1 \cos 4x + C_2 \cancel{\cos 4x} \frac{\sin 4x}{\cancel{\cos 4x}}$$

$$y = C_1 \cos 4x + C_2 \sin 4x$$

$$(1-2x-x^2)y'' + 2(1+x)y' - 2y = 0$$

$$y_1 = x+1$$

$$y'' + \frac{(2+2x)y'}{1-2x-x^2} - \frac{2y}{1-2x-x^2}$$

$$\text{here } P(x) = \frac{2(1+x)}{1-2x-x^2}$$

$$= - \int \frac{2+2x}{1-2x-x^2}$$

$$= + \int \frac{-2+2x}{1-2x-x^2}$$

$$= \int \ln|1-2x-x^2|$$

$$y_2 = u = \int \frac{\ln|1-2x-x^2|}{(x+1)^2} dx$$

$$u = \int \frac{1-2x-x^2}{(x+1)^2} dx$$

$$u = - \int \frac{x^2+2x-1}{(x+1)^2} dx$$

$$u = - \int \frac{x^2+2x-1}{x^2+2x+1}$$

$$u = - \int \frac{x^2+2x+1-2}{x^2+2x+1}$$

$$u = - \int x^2 \left(1 - \frac{2}{(x+1)^2} \right) dx \quad \begin{matrix} x^2 \\ x^2 \end{matrix}$$

$$u = - \int 1 dx - 2 \int \frac{1}{(x+1)^2} dx$$

Ex:- 4.2 (1-20)

4.3, 4.4, 4.6, 4.7, 4.9

$$u_{y_2} = \cancel{(2x+1)} \left(x - \frac{2(x+1)^{-2+1}}{-2+1} \right)$$

$$y_2 = -(x+1) \left(x + \frac{2}{(x+1)} \right)$$

$$y_2 = -x(x+1) + 2$$

$$y_2 = x^2 + x + 2$$

$$y = C_1(x+1) + C_2(x^2 + x + 2)$$

Lecture no 24

Homogenous Linear Differential Equations with constant

Co-efficients :

$$a_n y^{(n)} + a_{n-1} y^{(n-1)} + a_{n-2} y^{(n-2)} + \dots + a_2 y'' + a_1 y' + a_0 y = 0$$

This n^{th} order linear homogenous D.E where $a_0, a_1, a_2, \dots, a_n$ are real numbers.

examples:

$$4y'' + 3y = 0$$
$$y'' + 2y' - 3y = 0$$

$$\Rightarrow \text{If } m = \alpha \pm \beta i$$

$$y_c = e^{\alpha x} (C_1 \cos \beta x + C_2 \sin \beta x)$$

Q3

$$2y'' - 5y' - 3y = 0$$

Let $y = e^{mx}$
 $y' = me^{mx}$
 $y'' = m^2 e^{mx}$

eq(i) becomes

$$2m^2 e^{mx} - 5me^{mx} - 3e^{mx} = 0$$
$$e^{mx}(2m^2 - 5m - 3) = 0$$

$\therefore e^{mx}$ cannot be zero:-

$$2m^2 - 5m - 3 = 0 ; e^{mx} \neq 0$$

↓

Auxiliary equation

$$2m^2 - 6m + m - 3 = 0$$

$$2m(m-3) + 1(m-3) = 0$$

$$(2m+1)(m-3) = 0$$

$$m = 3 \quad \bigg/ \quad m = \frac{-1}{2}$$

$$y = C_1 e^{3x} + C_2 e^{-\frac{1}{2}x} \Rightarrow \underline{\text{General Solution}}$$

$\downarrow$
complementary solution

504

$$y'' - 10y' + 25y = 0$$

Auxiliary equation:-

$$m^2 - 10m + 25 = 0$$

$$m^2 - 5m - 5m + 25 = 0$$

$$m(m-5) - 5(m-5) = 0$$

$$(m-5)(m-5) = 0$$

$$(m-5)^2 = 0$$

$$m = 5 \quad / \quad m = 5$$

→ repeating roots

$$y_c = C_1 e^{5x} + C_2 x e^{5x}$$

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$$y'' + 4y' + 7y = 0$$

Auxiliary equation:-

$$m^2 + 4m + 7 = 0$$

$$m = \frac{-4 \pm \sqrt{16 - 28}}{2}$$

$$= \frac{-4 \pm \sqrt{-12}}{2}$$

$$= \frac{-4 \pm \sqrt{12} i}{2}$$

$$m = \frac{-4 \pm 2\sqrt{3}i}{2} \Rightarrow m = -2 \pm \sqrt{3}i$$

$$y_c = e^{-2x} (C_1 \cos \sqrt{3}x + C_2 \sin \sqrt{3}x)$$

$$\begin{array}{r} 2 \overline{) 12} \\ 2 \overline{) 6} \\ 3 \overline{) 3} \\ 1 \end{array}$$

Ex: 4.3 (1-14)

$$y''' + 3y'' - 4y = 0$$

Auxiliary equation

$$m^3 + 3m^2 - 4 = 0$$

$m=1$ use synthetic division :-

1	1	3	0	-4
1	↓	1	4	4
	1	4	4	0

$$m^2 + 4m + 4 = 0$$

$$m^2 + 2m + 2m + 4 = 0$$

$$m(m+2) + 2(m+2)$$

$$(m+2)(m+2) = 0$$

$$m = -2, -2$$

$$y_c = C_1 e^{-2x} + C_2 x e^{-2x} + C_3 e^x$$

$$y_1 = e^{-2x}, y_2 = x e^{-2x}, y_3 = e^x$$

↓
Superposition principle

{Q.17}

$$y''' - 5y'' + 3y' + 9y = 0$$

Auxiliary Equation

$$m^3 - 5m^2 + 3m + 9 = 0$$

$$[m=3]$$

$$\begin{array}{r|rrrr} & 1 & -5 & 3 & 9 \\ 3 & \downarrow & 3 & -6 & -9 \\ \hline & 1 & -2 & -3 & 0 \end{array}$$

$$m^2 - 2m - 3 = 0$$

$$m^2 - 3m + m - 3 = 0$$

$$m(m-3) + 1(m-3) = 0$$

$$(m+1)(m-3) = 0$$

$$m = -1 \quad | \quad m = 3$$

$$y_c = c_1 e^{-x} + c_2 e^{3x} + c_3 x e^{3x}$$

{Q.32}

$$4y'' - 4y' - 3y = 0$$

$$\begin{aligned} y(0) &= 1 \\ y'(0) &= 5 \end{aligned}$$

$$4m^2 - 4m - 3 = 0$$

$$4m^2 - 6m + 2m - 3 = 0$$

$$2m(2m-3) + 1(2m-3) = 0$$

$$(2m+1)(2m-3) = 0$$

$$m = -\frac{1}{2} \quad | \quad m = \frac{3}{2}$$

$$y_c = c_1 e^{-1/2x} + c_2 e^{3/2x}$$

$$y'_c = \frac{c_2 3}{2} e^{3/2x} - \frac{1}{2} c_1 e^{-1/2x}$$

Question will be with boundary value + initial value

Using initial condition :- Ex 4.3 (1-40)
(49-58)

when $x=0$ then $y=1$

$$1 = C_1 + C_2 \quad \text{--- (i)} \Rightarrow 1 - C_1 = C_2 \quad \text{--- (ii)}$$

when $x=0, y'=5$

$$5 = \frac{3}{2} C_1 - \frac{1}{2} C_2$$

$$5 = \frac{3}{2} C_1 - \frac{1}{2} (1 - C_1)$$

$$5 = \frac{3}{2} C_1 - \frac{1}{2} + \frac{1}{2} C_1$$

$$5 = \frac{4}{2} C_1 - \frac{1}{2}$$

$$5 = \frac{4C_1 - 1}{2}$$

$$10 = 4C_1 - 1$$

$$11 = 4C_1$$

$$\frac{11}{4} = C_1 \quad \text{put in (ii)}$$

$$1 - \frac{11}{4} = C_2$$

$$\frac{4 - 11}{4} = C_2$$

$$\frac{-7}{4} = C_2$$

$$y = \frac{11}{4} e^{3/2} - 7e^{-1/2} x$$

↓
general solution

{Q.49}

$$y = c_1 e^x + c_2 e^{5x}$$

$$m = 1, 5$$

$$(m-1)(m-5) = 0$$

$$m^2 - 6m + 5 = 0$$

$$y'' + 6y' + 5y = 0$$

$$y'' - 5y' + 6y = 0$$

$$y'' - 6y' + 5y = 0$$

$$y'' + 4y' - 2y = 2x^2 - 3x + 6$$

$g(x)$

$$y = y_c + y_p$$

complementary

solution $\Rightarrow y'' + 4y' - 2y = 0 \quad \text{--- (i)}$

Auxiliary equation is

let $y = e^{mx}$ be solution then
auxiliary eq is:-

$$m^2 + 4m - 2 = 0$$

$$m = -2 \pm \sqrt{6}$$

$$y_c = C_1 e^{(-2+\sqrt{6})x} + C_2 e^{(-2-\sqrt{6})x}$$

For y_p :- let $y = Ax^2 + Bx + C$. $\rightarrow$ Let max power $g(x)$

$$y' = 2Ax + B$$

$$y'' = 2A$$

put values of single derivative, y'' and y' in (i)

~~$$2A + 2Ax + B + Ax^2 + Bx + C = 0$$~~

$$2A + 4(2Ax + B) - 2(Ax^2 + Bx + C) = 2x^2 - 3x + 6$$

$$8A + 8Ax + 4B - 2Ax^2 - 2Bx - C = 2x^2 - 3x + 6$$

$$-2Ax^2 + 8Ax - 2Bx + 2A + 4B - C = 2x^2 - 3x + 6$$

$$2Ax^2 - 8Ax + 2Bx - 2A - 4B + C = -2x^2 + 3x - 6$$

$$x^2(2A) + x(8A) + x(2B) - (2A + 4B - C) = -2x^2 + 3x - 6$$

$$x^3: Ax^3 + Bx^2 + Cx + D$$

$G(x) :-$

if

$$g(x) = 2x - 7 \quad \text{then } y_p = Ax + B$$

$$g(x) = x^2 \quad \text{then } y_p = Ax^2 + Bx + C$$

$$g(x) = 6 \quad \text{" } y_p = A$$

if :-

$$g(x) = \sin x \text{ or } \cos x$$

$$\text{then } y_p = A \sin \alpha x + B \cos \alpha x$$

Reason

finding y

SOL

$$y'' + 4y = 3 \sin 2x$$

For y_c Auxiliary equation :-

$$m^2 + 4 = 0$$

$$m = \pm 2i$$

$$y_c = e^{0x} (C_1 \cos 2x + C_2 \sin 2x)$$

$$y_c = C_1 \cos 2x + C_2 \sin 2x$$

$$\therefore m = \alpha \pm \beta i$$

$$y_c = e^{\alpha x} (C_1 \cos \beta x + C_2 \sin \beta x)$$

Now for y_p we have let y
 $y = A \cos 2x + B \sin 2x$

method

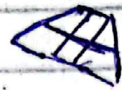
$$y = A \cos 2x + B \sin 2x$$

$$y' = -A \sin 2x (2) + B \cos 2x (2)$$

$$= -2A \sin 2x + 2B \cos 2x$$

$$y'' + 4y$$

$$y'' = -4A \cos 2x - 4B \sin 2x$$



~~$$-4A \cos 2x - 4B \sin 2x + 4(A \cos 2x + B \sin 2x) = 2x^2 - 3x + 6$$~~
~~$$-4A \cos 2x - 4B \sin 2x + 4A \cos 2x + 4B \sin 2x = 2x^2 - 3x + 6$$~~

$$-4A \cos 2x - 4B \sin 2x + 4(A \cos 2x + B \sin 2x) = 2x^2 - 3x + 6$$

$$0 = 2x^2 - 3x + 6$$

if y_p matches with y_c
 $\rightarrow$

Now let $y_p \Rightarrow$ let $y = (A \cos 2x + B \sin 2x) x$

$\downarrow$
 multiply by itself

$$y' = (A \cos 2x + B \sin 2x) + x(-2A \sin 2x + 2B \cos 2x)$$

$$y'' = -2A \sin 2x + 2B \cos 2x + (-2A \sin 2x + 2B \cos 2x) + x(-4A \cos 4x - 4B \sin 2x)$$

$$y'' = -4A \sin 2x + 4B \cos 2x - 4Ax \cos 2x - 4Bx \sin 2x$$

~~$$-4A \sin 2x + 4B \cos 2x - 4Ax \cos 2x - 4Bx \sin 2x$$~~
~~$$+ 4A \cos 2x + 4B \sin 2x + 4Ax \cos 2x + 4Bx \sin 2x = 3 \sin 3x$$~~

$$-4A \sin 2x + 4B \cos 2x - 4Ax \cos 2x - 4Bx \sin 2x + 4A \cos 2x + 4B \sin 2x + 4Ax \cos 2x + 4Bx \sin 2x = 3 \sin 3x$$

$$-4A = 3 \quad 4B = 0$$

$$A = -3/4 \quad B = 0$$

$$y_p = -\frac{3}{4} x \cos 2x$$

$$y_p = -\frac{3}{4} x \cos 2x$$

$$y' = -\frac{3}{4} (x \cos 2x)$$

$$= -\frac{3}{4} \{ x 2 \cos 2x + (\cos 2x) \}$$

$$y' = -\frac{3}{4} \{ -2x \sin 2x + \cos 2x \}$$

$$y'' = -\frac{3}{4} \{ -2 (x \sin 2x)' + (\cos 2x)' \}$$

$$y'' = -\frac{3}{4} \{ -2 (x \cos 2x + \sin 2x) - 2 \sin 2x \}$$

$$y'' = -\frac{3}{4} \{ -4x \cos 2x - 2 \sin 2x - 2 \sin 2x \}$$

$$= \frac{12x \cos 2x}{4} + \frac{6 \sin 2x}{4} + \frac{6 \sin 2x}{4}$$

$$y'' = 3 \cos 2x + \frac{3 \sin 2x}{2} + \frac{3 \sin 2x}{2}$$

$$= 3 \cos 2x + \frac{3 \sin 2x}{2} + \frac{3 \sin 2x}{2} + 4 \left(-\frac{3}{4} \right) x \cos 2x$$

$$= 3 \cancel{\cos 2x} + \frac{3 \sin 2x}{2} + \frac{3 \sin 2x}{2} - 3 \cancel{\cos 2x} = 3 \sin 2x$$

$$= \cancel{2 \sin 2x} + \frac{6 \sin 2x}{2} = 3 \sin 2x$$

$$6 = 3$$

$$2$$

$$\boxed{6 = 6}$$

$G(x)$

Ex: 4.4 (1-40)

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Table

Ex: 6

2 conditions Superposition Rule

if $g(x) = 2x \sin 3x$

$$y_p = (Ax+B) \sin 3x + (Cx+D) \cos 3x$$

if $g(x) = 2x^2 \cos x \Rightarrow$ not in exam

$$y_p = (Ax^2+Bx+C) \cos x + (Dx^2+Ex+F) \sin x$$

if $g(x) = e^{-2x} \Rightarrow$ $7e^{-2x}$
 $y_p = Ae^{-2x}$

$U(x) =$ method
L or
 $\sin, \cos x$
Multiplication
Polynomial
~~if~~

$$g(x) = (2x-7)e^{-2x}$$
$$y_p = (Ax+B)e^{-2x}$$

$$y_p = (Ax+B)e^{-2x}$$

$$y_p = Ax e^{-2x} + Be^{-2x}$$

$$y_c = c_1 e^{-2x} + c_2 e^{3x}$$

match so multiply y_p with x

Cauchy-Euler Equation:

Example:-

$$2x^2 y'' - 6xy' + 4y = 6 \sin x$$

↓
variable coefficient equation

For ↓ in this (this is the difference)

{0}

$$x^2 y'' - 2xy' - 4y = 0$$

For y_c , let $y = x^m \neq 0$

$$y' = mx^{m-1}$$

$$y'' = (m-1)m(x)^{m-2}$$

put (i) and (ii)

$$x^2 (m^2 - m)x^{m-2} - 2xm x^{m-1} - 4x^m = 0$$

$$x^m (m^2 - m) - 2mx^m - 4x^m = 0$$

$$x^m [m^2 - m - 2m - 4] = 0$$

$$x^m [m^2 - 3m - 4] = 0$$

$$x^m \neq 0 \text{ so } m^2 - 3m - 4 = 0$$

$$m^2 - 4m + m - 4 = 0$$

$$m(m-4) + 1(m-4) = 0$$

$$(m+1)(m-4) = 0$$

$$m = -1, 4$$

$$y_c = C_1 x^{-1} + C_2 x^4$$

{Q}

$$4x^2y'' + 8xy' + y = 0 \Rightarrow \text{homogenous}$$

For y_c , let $y = x^m$, the auxiliary equation in m is given by :-

$$4m(m-1) + 8m + 1 = 0$$

$$4m^2 - 4m + 8m + 1 = 0$$

$$4m^2 + 4m + 1 = 0$$

$$m = \frac{-1}{2}, m = \frac{-1}{2} \text{ for repeating multiply}$$

$$y = x^m \text{ with } \ln x$$

$$y_c = C_1 x^{-1/2} + C_2 (\ln x) x^{-1/2}$$

TABLE

if $y = x^a (C_1 \cos(b \ln x) + C_2 \sin(b \ln x))$

$$y_c = x^a (C_1 \cos(b \ln x) + C_2 \sin(b \ln x)) \rightarrow m = a \pm i b$$

$$y_c = C_1 \cos(\ln x) + C_2 \sin(\ln x)$$

$$\rightarrow m = \pm i$$

$$y_c = C_1 x^{-1} + C_2 x$$

$$\pm 1$$

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$$x^2 y'' + xy' - y = \ln x$$

$$m(m-1) + m - 1 = 0$$

$$m^2 - m + m - 1 = 0$$

$$m^2 - 1 = 0$$

$$m^2 = 1$$

$$m = \pm 1$$

$$y_c = C_1 x + C_2 x^{-1}$$

$$y_c = C_1 x + C_2 x^{-1}$$

$$f(x) = \frac{\ln x}{x^2}$$

↓
for Wronskian
determinant

$$y_p = u_1 y_1 + u_2 y_2$$

$$u_1 = \int \frac{w_1}{w} dx, \quad u_2 = \int \frac{w_2}{w} dx$$

↓

$$w = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} x & 1/x \\ 1 & -1/x^2 \end{vmatrix} = \frac{-1}{x} - \frac{1}{x} = \frac{-2}{x} = w$$

$$w_1 = \begin{vmatrix} 0 & y_2 \\ f(x) & y_2' \end{vmatrix} = \begin{vmatrix} 0 & 1/x \\ 1/x + \ln x & -1/x^2 \end{vmatrix}$$

$$w_1 = \frac{-1}{x^2} \ln x$$

$$w_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f(x) \end{vmatrix} = \begin{vmatrix} x & 0 \\ 1 & \frac{1}{x^2} \ln x \end{vmatrix}$$

$$= \frac{1}{x} \ln x$$

$$u_1 = \int \frac{-\frac{1}{x^3} \ln x}{-\frac{2}{x}}$$

$$u_1 = \int \frac{-1}{x^3} \ln x \times \frac{x}{-2}$$

$$u_1 = \int \frac{-1}{-2x^2} \ln x (dx)$$

$$u_1 = \frac{1}{2} \int \frac{\ln x}{x^2} dx$$

$$u_2 = \int \frac{\frac{1}{x} \ln x}{-\frac{2}{x}} dx$$

$$u_2 = \frac{-1}{2} \int \ln x dx$$